Retail Customer Intelligence & Sales Analytics System

Project Overview

This project analyzes customer shopping data to understand purchasing behavior and identify valuable business insights. Using **Python, SQL, Excel, and Power BI**, the project explores customer demographics, product preferences, shopping patterns, and sales performance to help businesses make informed decisions.

Dataset Summary

Dataset Information

- **Total Records:** 3,900
- **Total Features:** 18

Dataset Includes

- Customer Information (Age, Gender, Location, Subscription Status)
- Product Information (Item Purchased, Category, Size, Color)
- Purchase Details (Purchase Amount, Season, Payment Method)
- Shopping Behavior (Discount Applied, Review Rating, Shipping Type, Previous Purchases)

Data Quality

- Missing values were found in the **Review Rating** column and handled during data preprocessing.

Customer	Age	Gender	Item Purchased	Category	Purchase Amount	Location	Size	Color	Season	Review Rating	Subscription	Shipping Type	Discount Applied	Promo Code	Previous Purchases	Payment Method	Frequency of Purchases
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
3	50	Male	Jeans	Clothing	73	Massachus	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
4	21	Male	Sandals	Footwear	90	Rhode Isla	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash	Quarterly
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly
9	26	Male	Coat	Outerwear	97	West Virgi	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually
10	57	Male	Handbag	Accessorie	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	Cash	Quarterly
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	Yes	26	Bank Transfer	Bi-Weekly
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	Yes	10	Bank Transfer	Fortnightly
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Venmo	Fortnightly
14	65	Male	Dress	Clothing	51	New Hamp	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	PayPal	Weekly
15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	Yes	34	Debit Card	Weekly

Data Preparation & Analysis (Python)

The dataset was cleaned and prepared using Python.

Tasks Performed

- Imported the dataset using Pandas.
- Explored the dataset using info(), describe(), and head().

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- Identified and handled missing values.
- Standardized column names for better readability.
- Created new features such as **Age Group**.
- Removed unnecessary or duplicate information.
- Connected the cleaned dataset to PostgreSQL/MySQL for further analysis.

For PostgreSQL:

```
##### Connecting Python script to PostgreSQL
```

```
pip install psycopg2-binary sqlalchemy
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: psycopg2-binary in c:\users\pratham\appdata\roaming\python\python313\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in c:\programdata\anaconda3\lib\site-packages (2.0.43)
Requirement already satisfied: greenlet>=1 in c:\programdata\anaconda3\lib\site-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\programdata\anaconda3\lib\site-packages (from sqlalchemy) (4.15.0)
Note: you may need to restart the kernel to use updated packages.
```

```python
from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
# Replace placeholders with your actual details
username = "postgres"      # default user
password = "123123123" # the password you set during installation
host = "localhost"         # if running locally
port = "5432"              # default PostgreSQL port
database = "customer_behavior"    # the database you created in pgAdmin

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer"   # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

For MySQL:

```
### Code for MySQL
```

```python
!pip install pymysql sqlalchemy
from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "your_password"
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

# Write DataFrame to MySQL
table_name = "customer"   # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

# Read back sample
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

Business Analysis (SQL)

SQL was used to answer important business questions and generate meaningful insights.

Analysis Performed

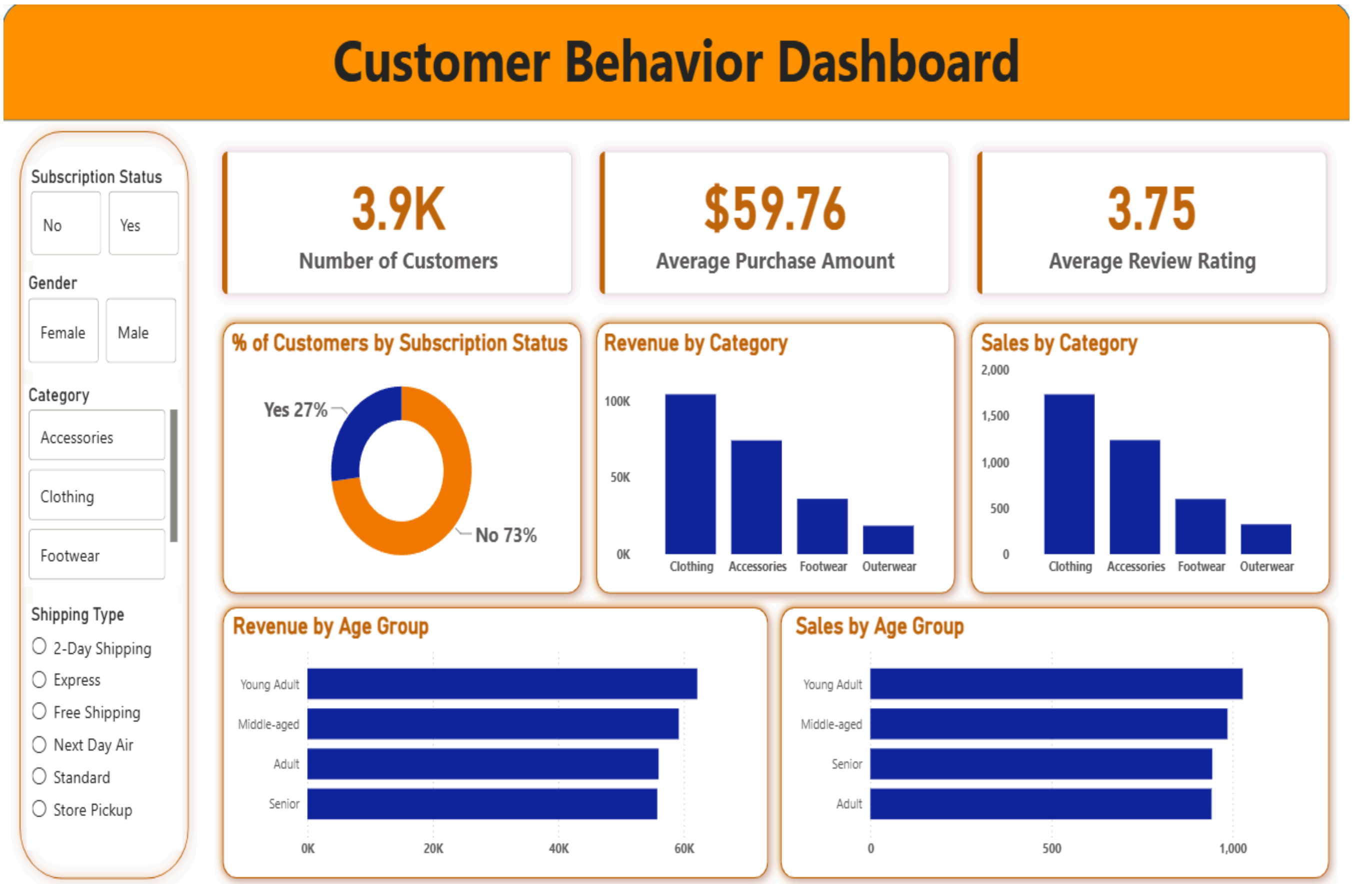
- Compared revenue generated by male and female customers.
- Identified customers who spent more while using discounts.
- Found the highest-rated products.
- Compared purchase amounts across different shipping methods.
- Analyzed spending patterns of subscribers and non-subscribers.
- Identified products with the highest discount usage.
- Segmented customers into New, Returning, and Loyal groups.
- Found the most purchased products in each category.
- Analyzed repeat customers and subscription behavior.
- Calculated revenue generated by different age groups.

Dashboard Development (Power BI)

An interactive dashboard was developed to present business insights in a simple and visually appealing way.

Dashboard Highlights

- Sales Overview
- Customer Demographics
- Product Category Performance
- Revenue Analysis
- Subscription Insights
- Discount Analysis
- Customer Segmentation
- Interactive Filters and KPIs



Key Business Insights

The analysis revealed several useful insights, including:

- Customer purchasing behavior differs across age groups and product categories.
- Subscription members contribute significantly to overall sales.
- Discounts influence purchasing decisions for specific products.
- Some products consistently receive higher customer ratings.
- Repeat customers play an important role in business growth.

Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Introduce attractive loyalty programs to encourage repeat purchases.
- Offer personalized promotions to different customer segments.
- Promote high-performing and top-rated products.
- Review discount strategies to maximize both sales and profit.
- Use customer insights to improve marketing campaigns and customer experience.

Conclusion

The **Retail Customer Intelligence & Sales Analytics System** demonstrates how customer shopping data can be transformed into meaningful business insights using **Python, SQL, Excel, and Power BI**. The project highlights the importance of data-driven decision-making and provides practical recommendations that can help retail businesses improve customer satisfaction, increase revenue, and support long-term business growth.